



**ITSY 2372
Ethical Hacking**

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Determining Vulnerabilities of Wireless Networks

Submitted to
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by

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1. Executive Summary

A wireless network security assessment was done on the Alexander Rocco Network. It has been discovered that the wireless router is configured to issue IP addresses to connecting stations. It is also indicated by Vistumbler that channel 6 is active, the SSID is linksys, and WEP is enabled. Based on this, it suggests that there are many vulnerabilities that may be exploited, there are a handful of recommendations that can be made to mitigate these risks.

2. Findings

Based on the discoveries, it may be suggested that a few key issues are the use of WEP encryption, its default SSID "linksys", and the use of channel 6. The use of WEP can be easily cracked with a variety of tools such as Wireshark or Kismet. The default SSID "linksys" indicates that the router may still be using default configurations. Lastly the use of channel 6 may create interference and congestion, which may slow down performance.

3. Recommendations

A few recommendations for changes and upgrades to be made are to upgrade to WPA3 or WPA3 encryption, change the default SSID, update the router login, disable WPS, or even restrict DHCP and monitor devices. Doing these things increases wireless security while limiting the number of approved devices.

4. Conclusion

The Alexander Rocco Company's current wireless configuration have a high potential risk to its integrity and security of its internal network. Using a default SSID, unsecure DHCP, and WEP leaves it too exposed, but through a few changes, the company's security may significantly increase and reduce risks.